EfficientNet-B0 Architecture for Privacy-Preserving Brain Tumor Classification

A technical explanation, architecture diagram and project integration guide

Project: Privacy-Preserving Federated Learning Framework for Early Brain Tumor Classification Using Multi-Hospital MRI Images

Classification: Glioma, Meningioma, Pituitary and No Tumor

Architecture: EfficientNet-B0 (often mistyped as EfficientBO; the correct designation uses the digit zero).

Executive overview

EfficientNet-B0 is a convolutional neural network designed to provide a useful balance of image-classification accuracy and computational efficiency. It uses mobile inverted bottleneck convolution (MBConv), squeeze-and-excitation attention, and compound scaling principles. In this project, the model processes brain MRI images locally at simulated hospitals; a federated server combines trained model parameters without receiving the raw images.

Research context

The supplied 2025 article by Muntaqim and Smrity evaluates VGG19 and other CNN backbones under non-IID federated learning using FedAvg, FedProx and SCAFFOLD. It does not report EfficientNet-B0 results. Replacing or comparing VGG19 with EfficientNet-B0 is a proposed project advancement, not a reproduced result from that paper.

Main architectural diagram

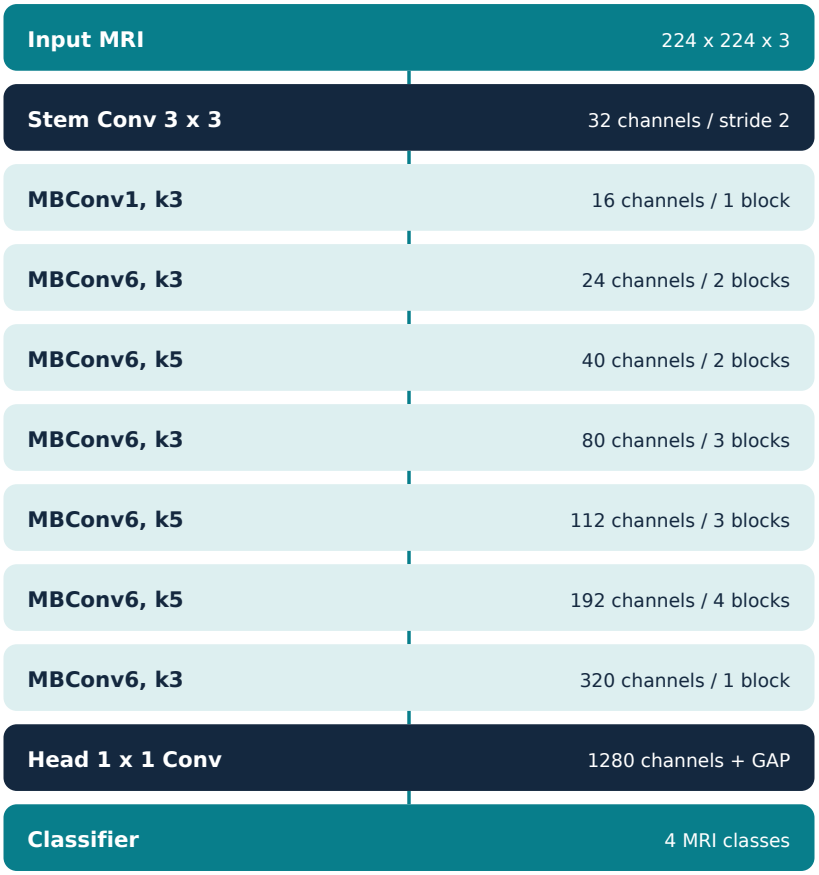


Diagram is a simplified stage-level illustration of the canonical EfficientNet-B0 backbone; spatial dimensions and classifier parameters depend on implementation.

1. How EfficientNet-B0 works

The input MRI image is resized to 224 × 224 pixels and converted to three channels when using ImageNet-pretrained weights. A stem convolution extracts basic edges and textures. Subsequent MBConv blocks learn increasingly abstract image patterns, such as tissue boundaries and spatial structures. A final convolution and global average pooling generate compact features, followed by a new four-output classifier.

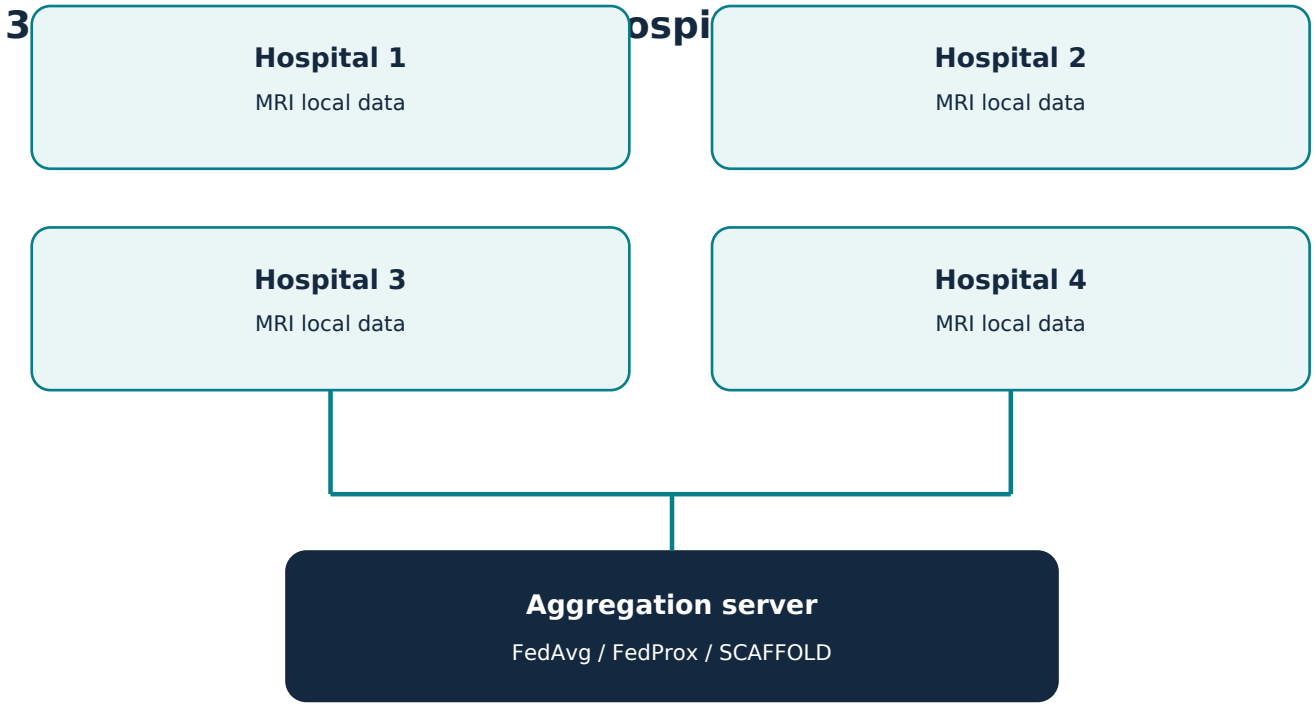
Key building blocks

Component	Purpose in the MRI classifier
Stem convolution	Initial edge, contrast and texture extraction
MBConv1 / MBConv6	Inverted residual blocks; expand channels, perform efficient depthwise convolution, then project features
Squeeze-and-excitation (SE)	Learns channel-wise weights to emphasize useful feature responses
Swish / SiLU activation	Smooth nonlinear activation used in EfficientNet blocks
Global average pooling	Converts spatial feature maps into a compact feature vector
Four-class classification head	Produces logits and softmax probabilities for glioma, meningioma, no tumor and pituitary

2. Architecture stages and approximate parameters

Stage	Operator	Output channels	Repetitions
1	3×3 convolution	32	1
2	MBConv1, 3×3	16	1
3	MBConv6, 3×3	24	2
4	MBConv6, 5×5	40	2
5	MBConv6, 3×3	80	3
6	MBConv6, 5×5	112	3
7	MBConv6, 5×5	192	4
8	MBConv6, 3×3	320	1
9	1×1 convolution + pooling	1280	1
10	Task-specific dense layer	4	1

EfficientNet-B0 has approximately 5.3 million parameters in its standard ImageNet 1000-class configuration. Replacing the classifier with a four-class head changes the total. The exact count must be printed from the actual PyTorch model.



Global EfficientNet-B0 weights return to clients each round

The figure shows four simulated clients and a shared aggregation server. MRI images remain in each client partition; only model updates are exchanged in the basic simulation. This alone is not a guarantee of privacy against model-update attacks.

Federated learning workflow

1. Divide the training data into four disjoint hospital-client partitions. Use both IID and non-IID class distributions, preserving a separate untouched test set.

2. Initialize one global EfficientNet-B0 with a four-class classification head. Send identical initial weights to the clients.

3. Each client trains locally on its MRI subset for a specified number of epochs and returns updated parameters or parameter differences.

4. Aggregate updates at the server using FedAvg, FedProx or SCAFFOLD. FedProx modifies the local objective; SCAFFOLD uses control variates to correct client drift, so these methods are not merely three interchangeable averaging formulas.

5. Redistribute the global model and repeat the federated rounds. Evaluate on validation data during development; use the untouched test set for final reporting.

Why EfficientNet-B0 can help

Its relatively compact backbone may reduce local memory use, computation and communication compared with larger VGG architectures when the entire backbone is transmitted. It can therefore be a practical candidate for simulated multi-hospital federated learning. However, actual speed, accuracy, communication bytes and stability must be benchmarked in your Colab environment.

4. Recommended preprocessing

Use image decoding, RGB conversion, resizing to 224 × 224, and the exact normalization expected by the selected pretrained weights. Augmentation should be applied only to training images. Maintain consistent class-to-index mapping during training and inference. Check duplicate images and, when identifiers exist, split by patient rather than by image to reduce leakage.

5. Grad-CAM and MRI upload AI Studio

Grad-CAM produces a heatmap of image regions that influenced a selected model prediction. It is an explanation of classifier attention, not a verified tumor segmentation or diagnostic localization. For an EfficientNet-B0 PyTorch model, register hooks on a suitable final spatial convolution feature layer; run a forward pass with gradients enabled, backpropagate the selected class score, and combine activation maps with pooled gradients.

Correct prediction pipeline

Step	Expected behavior / common failure prevention
Upload	Accept PNG/JPEG MRI; validate that an image was decoded
Preprocess	Apply the same resize, RGB conversion and normalization used for training
Load model	Rebuild the same architecture and class order, then load the trained global checkpoint
Inference	Run evaluation mode; compute logits and softmax class probabilities
Grad-CAM	Enable gradients for the selected class; capture a spatial feature map; normalize and overlay heatmap
Results	Display predicted class, confidence, all four probabilities, original image and Grad-CAM overlay

Example interpretation

If the uploaded MRI is classified as glioma with a confidence score, the web application can display the predicted class and a Grad-CAM overlay indicating influential regions. A highlighted area does not independently confirm the presence, type or boundaries of a tumor. A radiologist and appropriately validated clinical workflow are required for diagnosis.

6. Comparison plan for project review

Model	Research role	What to compare
VGG16	Original base-paper baseline	Accuracy, macro-F1, parameter count
VGG19	2025 non-IID article baseline	Accuracy, convergence, calibration
ResNet-18	Compact residual CNN comparison	Inference time and robustness
EfficientNet-B0	Proposed main architecture	Accuracy, efficiency and non-IID stability

For each architecture, compare FedAvg, FedProx and SCAFFOLD under matched IID/non-IID partitions, random seeds, training budgets and test data. Report confidence intervals and paired tests across repeated runs. Do not treat the article’s reported VGG19 accuracy as your own result.

7. Advantages, limitations and research contributions

Potential advantages

EfficientNet-B0 provides a compact pretrained backbone, MBConv-based feature extraction, SE channel attention and an accessible transfer-learning starting point. These properties make it a reasonable candidate for lower-resource federated experiments.

Limitations and responsible use

Performance may degrade under severe non-IID label skew, dataset shift, low-quality MRI, small client datasets or mismatched preprocessing. Four partitions of a public Kaggle dataset are simulated hospitals, not four independently collected clinical institutions. Federated learning does not automatically provide differential privacy, secure aggregation or resistance to information leakage. Grad-CAM is not segmentation. This system is a research prototype, not a clinically validated diagnostic tool.

Suggested measurable research contributions

A fair four-backbone comparison; controlled IID versus non-IID stress tests; FedAvg/FedProx/SCAFFOLD ablations; parameter and communication-cost measurements; calibration analysis; cross-dataset evaluation with compatible labels; reproducible multi-seed statistics; and an MRI upload demo that verifies model-checkpoint and preprocessing consistency.

8. Viva / panel explanation

Why EfficientNet-B0? It is a compact CNN that combines MBConv blocks, squeeze-and-excitation and an efficient scaling design. We test whether it offers a better accuracy-efficiency trade-off than VGG and ResNet baselines.

What is its role in federated learning? It is the local and global classification backbone. Hospitals train locally; the server aggregates model updates across rounds.

What does Grad-CAM show? Image regions influencing the model prediction, not clinically confirmed tumor boundaries.

What proves an improvement? Matched experimental results on held-out data, repeated runs, confidence intervals and resource measurements.

References and provenance

1. Tan, M. and Le, Q. V. (2019). EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks. ICML. https://arxiv.org/abs/1905.11946

2. Muntaqim, M. Z. and Smrity, T. A. (2025). Federated Learning Framework for Brain Tumor Detection Using MRI Images in Non-IID Data Distributions. Journal of Imaging Informatics in Medicine, 38, 3909–3927. DOI: 10.1007/s10278-025-01484-9. Supplied research article.

3. Project-specific discussion is an engineering proposal informed by the user’s federated learning notebook and project requirements; it is not a claim that the cited paper implemented EfficientNet-B0.